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Contents

Chapter 1

We gotta start somewhere

In this thesis we will be studying the Vicsek set, so therefore we will in this section start by constructing the set and proving some basic properties of the Vicsek set.

1.1 Construction the Vicsek set

Let $V_0 = \{p_1, p_2, p_3, p_4, p_5\}$ be the set of vertices in a plus with a middle point in $\mathbb{R}^2$. Then define

$$\mathcal{S}_i(z) = \frac{1}{3}(z - p_i) + p_i, \quad \text{for } i = 1, 2, \ldots 5.$$

The Vicsek set K is then the unique non-empty compact subset in $\mathbb{C}$ such that

$$K = \bigcup_{i=1}^{5} \mathcal{S}_i(K)$$

V_0 is the boundary of K.

Next we can then define a sequence of sets $(V_m)_{m\geq 0}$ inductively:

$$V_{m+1} = \bigcup_{i=1}^{5} \mathcal{S}_i(V_m)$$

this gives us a sequence of Vicsek set graphs $(G_m)_{m\geq 0}$ whose vertices are $(V_m)_{m\geq 0}$. Further we will use the notation of $V_* = \cup_{m\geq 0} V_m$ and for any $p \in V_m$ we denote the collection of neighbours of p in G_m by $V_{m,p}$. Lastly note that $N_m = \#V_m = 5^m \cdot 4 + 1$.

1.2 Dirichlet Forms

In this section we will follow arguments given by [Str06].

In this section we wish to introduce the notion of Dirichlet forms. This will allow us to introduce the Dirichlet Laplacian and an important constant. Let us start by introducing the Dirichlet form.

Definition 1.1: Let V be a finite set. A symmetric bilinear form on $\ell(V)$

the standard graph Dirichlet form.

Definition 1.2: Given a finite connected graph K and it vertices V_m the for a function $u : V_m \to \mathbb{R}$ we define the *standard graph Dirichlet form* to be

$$E_K(u) = \sum_{x\sim y} \big(u(x) - u(y)\big)^2 \tag{1.1}$$

Proposition 1.3: Equation (1.1) is indeed a Dirichlet form

Proof. obvious? □

Given some graph K and a function $u^{(m)}: V_m \to \mathbb{R}$, let's say we want to extend this function to V_{m+1}. This leads us to defining an extension in the following way

Definition 1.4: Let K be a graph and $u: V_m \to \mathbb{R}$, then an *extension of* u is a function $u': V_{m+n} \to \mathbb{R}$ where $u'|_{V_m} = u$.

It is obvious that $E_{m+n}(u') \geq E_m(u)$. Further one quickly notices that there must be a function that minimizes the energy of a given graph. We denote such a function with $\tilde{u}$ and call it a *harmonic extension*.

Definition 1.5: Let $\tilde{u}: V_{m+n} \to \mathbb{R}$ with $|_{V_m} = u$ is called a *harmonic extension* if $E_{m+n}(\tilde{u}) \leq E_{m+n}(u')$ for all $u': V_{m+n} \to \mathbb{R}$ with $u'|_{V_m} = u$.

In some cases we have that $E_{m+n}(\tilde{u}) = rE_m(u)$ for some $r \in (0,1)$. But instead of normalizing the values on E_m we will normalize E_{m+n}, and getting

$$r^{-1}E_{m+n}(\tilde{u}) = E_m(u), \quad \text{and} \quad r^{-1}E_{m+n}(u') \geq E_m(u)$$

for every extension u' of u. From the equality above we can define the following

Definition 1.6: The *renormalized graph energies* are defined as

$$\mathcal{E}_m(u) = r^{-m}E_m(u)$$

Remark 1.7: The sequence $\{\mathcal{E}_m(u)\}$ is a non-decreasing sequence. Something with it being constant if u is linear and how is u defined? In the sequence $\{\mathcal{E}_m(u)\}$ if we have u defined on V_m what is $\mathcal{E}_{m+1}(u)$? Any extension or the harmonic extension?

Now that we have 1.6, we can define what it means to be a harmonic function.

Definition 1.8: u is a *harmonic function* if u minimizes $\mathcal{E}_m(u)$ for all m, given boundary values V_0.

Minimizing Energy and finding renormalization constant on the Vicsek set

I the previous section we have talked about a harmonic function, but without every constructing one. We will therefore in this next section show how to create an harmonic function for the Vicsek set for any start boundary values.

Given $V_0 = \{q_0, q_1, q_2, q_3, q_4\}$ and $u: V_0 \to \mathbb{R}$ with $u(q_i) = \alpha_i$ for $i = 1, 2, \ldots, 5$. Then we have that

$$E_0(u) = \big(u(q_0) - u(q_1)\big)^2 + \big(u(q_0) - u(q_2)\big)^2 + \big(u(q_0) - u(q_3)\big)^2 + \big(u(q_0) - u(q_4)\big)^2.$$

We are going to exploit symmetry of the Vicsek set. Since all four "limbs" of the Vicsek set are identical we will only look at one of the limbs since the other limbs will be the same computations. So if we want to minimize $E_1(u')$, we see for the limb between q_0 and q_1 that

$$E_1(u') = \big(u'(q_1) - u'(p_0)\big)^2 + \big(u'(p_0) - u'(p_2)\big)^2 + \big(u'(p_0) - u'(p_3)\big)^2 + \big(u'(p_0) - u'(p_4)\big)^2 + \big(u'(p_3) - u'(q_0)\big)^2$$

We are then going to optimize with regards to $u'(p_i)$ for $i = 0, 1, 2, 3$. Taking the derivative of the right side and solving for 0 we get the following

$$\begin{aligned} 4u'(p_0) &= u'(q_1) + u'(p_2) + u'(p_3) + u'(p_4) \\ 2u'(p_3) &= u'(q_0) + u'(p_0) \\ u'(p_2) &= u'(p_0) \\ u'(p_4) &= u'(p_0). \end{aligned}$$

Solving these equations we get

$$\begin{aligned} u'(p_0) &= \frac{2}{3}u'(q_1) + \frac{1}{3}u'(q_0) \\ u'(p_2) &= \frac{2}{3}u'(q_1) + \frac{1}{3}u'(q_0) \\ u'(p_4) &= \frac{2}{3}u'(q_1) + \frac{1}{3}u'(q_0) \\ u'(p_3) &= \frac{2}{3}u'(q_0) + \frac{1}{3}u'(q_1). \end{aligned}$$

So what we have found is that to minimize the energy of the Vicsek set, then the energy of the next point is going to be $2/3$ the energy from the closest point and $1/3$ the energy from the next closest point.

If one calculates the energies then you will get

$$E_1(u') = \frac{1}{3}\sum_{i=1}^{4}(u'(q_i) - u'(q_0))^2 = \frac{1}{3}E_0(u')$$

Since the Vicsek set is constructed from V_m to V_{m+1} by creating a "plus" between any two points that are neighbours in V_m. So going from V_1 to V_2 is the same as going from V_m to V_{m+1}. In which case

$$E_{m+1}(u') = \frac{1}{3}E_m(u')$$

Thus we get our renormalization constant to be $r = 3$

Lemma 1.9: Let $v : K \to \mathbb{R}$ be a harmonic function on the Vicsek set. If $v(x) = v(y_i) = 0$ for all $y_i \in V_{m,x}$ then $v(\omega) = 0$ for all $\omega \in \overline{B_x}$ where $B_x = \left(\bigcup_{|w|>0} F_w x\right) \cup \left(\bigcup_i \{y_i\}\right)$

Proof. It is clear that $v(\omega) = 0$ for any $\omega \in B_x$, since the energy of the points $F_w x$ with $|w| > 0$ depends on the energy of x and its neighbours.

So we need to check the points in $\overline{B_x} \backslash B_x$. Now since B_x is dense in $\overline{B_x}$, then for any $\omega \in \overline{B_x}$ we have that there exists a sequence $x_n \in B_x$ such that $x_n \to \omega$. By the fact that v is continuos on K it is continuos on $\overline{B_x}$, in which case $\left|v(x_n) - v(\omega)\right| \to 0$ as $n \to \infty$, but $v(x_n) = 0$ for all x_n, so $v(\omega) = 0$ as desired. $\square$

What lemma 1.9 sates, is if any two points in V_m are neighbours, say x and y, and $v(x) = v(y) = 0$, then then function value at any point between $x$and y is also zero.

Lemma 1.10: Set $F_m = \operatorname{supp} \psi_x^{(m)}$, then

$$F_0 \supset F_1 \supset F_2 \supset F_3 \supset \ldots$$

Further F_m is contained in the open ball $B(x, \epsilon_m)$ where $\epsilon_m \to 0$ as $m \to \infty$.

Proof. If we set $\Omega_m = \left\{z \in K : \psi_x^{(m)}(z) = \psi_x^{(m)}(y) = 0, \forall y \in V_{m,z}\right\}$ then $F_m = K \backslash \left(\bigcup_{\alpha \in \Omega_m} \overline{B_\alpha}\right)$. It is clear that $\Omega_m \subset \Omega_{m+1}$ and that there are new points in Ω_{m+1} that are not just a point between some already existing points $x_1, x_2 \in \Omega_m$. We therefore have from the way that F_m is constructed that $F_m \supset F_{m+1}$.

The containment of F_m in $B(x, \epsilon_m)$ follows from lemma 1.9 and that $F_m \supset F_{m+1}$. $\square$

Theorem 1.11: Let $u \in \operatorname{dom} \Delta_\mu$. Then the pointwise formula

$$\Delta_\mu u(x) = \lim_{m\to\infty} r^m \left(\int_K \psi_x^{(m)} \, \mathrm{d}\mu\right)^{-1} \delta_m u(x)$$

holds with the limit uniform across $V_* \backslash V_0$. Conversely, suppose u is a continuos function and the right side of the limit converges uniformly to a continuos function on $V_* \backslash V_0$. Then $u \in \operatorname{dom} \Delta_\mu$ and the equation holds.

Proof. Suppose $u \in \operatorname{dom} \Delta_\mu$ and $\Delta_\mu u = f$, then by [INSERT REF] we have that

$$r^{-m} \left(\int_K \psi_x^{(m)} \, \mathrm{d}\mu\right)^{-1} \Delta_m u(x) = \frac{\int_K f \psi_x^{(m)} \, \mathrm{d}\mu}{\int_K \psi_x^{(m)} \, \mathrm{d}\mu}$$

for any $x \in V_m \backslash V_0$. Uniform converges of the right side is because of continuity of $f(x)$, so let $\epsilon > 0$ be given, then by lemma 1.10 we can choose m large enough so for all $x, y \in F_m$ that $\left|f(x) - f(y)\right| < \epsilon$, then

$$\begin{aligned}\left|\frac{\int_K f\psi_x^{(m)}\ \mathrm{d}\mu}{\int_K \psi_x^{(m)}\ \mathrm{d}\mu} - f(x)\right| &= \left|\frac{\int_K f\psi_x^{(m)}\ \mathrm{d}\mu}{\int_K \psi_x^{(m)}\ \mathrm{d}\mu} - \frac{\int_K f(x)\psi_x^{(m)}\ \mathrm{d}\mu}{\int_K \psi_x^{(m)}\ \mathrm{d}\mu}\right| \\ &= \left|\frac{\int_K \left(f(y) - f(x)\right)\psi_x^{(m)}\ \mathrm{d}\mu}{\int_K \psi_x^{(m)}\ \mathrm{d}\mu}\right| \\ &\leq \frac{\int_{F_m} \left|f(y) - f(x)\right|\ \mathrm{d}\mu}{\int_K \psi_x^{(m)}\ \mathrm{d}\mu} \\ &\leq \frac{\epsilon \int_{F_m}\ \mathrm{d}\mu}{\int_K \psi_x^{(m)}\ \mathrm{d}\mu} \\ &= \epsilon\end{aligned}$$

in which case

$$\lim_{m\to\infty} r^{-m}\left(\int_K \psi_x^{(m)}\ \mathrm{d}\mu\right)^{-1}\Delta_m u(x) = \lim_{m\to\infty}\frac{\int_K f\psi_x^{(m)}\ \mathrm{d}\mu}{\int_K \psi_x^{(m)}\ \mathrm{d}\mu} = f(x) = \Delta_\mu u(x)$$

Moving on to the second half of the proof. Suppose that $r^m\left(\int_K \psi_x^{(m)}\ \mathrm{d}\mu\right)^{-1}\delta_m u(x)$ converges uniformly to a continuos function f. Now for any $v \in \operatorname{dom}_0 \mathcal{E}$ we have

$$\mathcal{E}_m(u, v) = r^{-m}\sum_{\substack{x \sim y \\ m}}\left(u(y) - u(x)\right)\left(v(y) - v(x)\right).$$

If we collect all the terms that contain the factor $v(x)$ for a fixed $x \in V_m \backslash V_0$ then each $v(x)$ will be of the form $-v(x)\left(u(y) - u(x)\right)$ for each $y \underset{m}{\sim} x$. In which case we get

$$\begin{aligned}\mathcal{E}_m(u, v) &= -r^{-m}\sum_{x\in V_m\backslash v_0} v(x)\left(\sum_{\substack{x \sim y \\ m}}\left(u(y) - u(x)\right)\right) \\ &= -\sum_{x\in V_m\backslash V_0} v(x) r^{-m}\Delta_m u(x)\end{aligned}$$

While it seems we lose many terms by removing $v(y)$, but they are instead contained in $\sum_{\substack{x \sim y \\ m}}\left(u(y) - u(x)\right)$. Note that the sum is "reset" for every x so we count both the relation $x \sim y$ and $y \sim x$.

Now if we define $f_m(x) = r^{-m}\left(\int_K \psi_x^{(m)}\ \mathrm{d}\mu\right)^{-1}\Delta_m u(x)$ then $f_m \to f$ uniformly by assumption and we can rewrite the above as

$$\begin{aligned}\mathcal{E}_m(u, v) &= -\sum_{x\in V_m\backslash V_0} v(x)\left(\int_K \psi_x^{(m)}\ \mathrm{d}\mu\right) r^{-m}\left(\int_K \psi_x^{(m)}\ \mathrm{d}\mu\right)^{-1}\Delta_m u(x) \\ &= -\int_K\left(\sum_{x\in V_m\backslash V_0} v(x) f_m(x)\psi_x^{(m)}\right)\ \mathrm{d}\mu.\end{aligned}$$

Following a quite lengthy argument, we have that

$$\mathcal{E}(u, v) = \lim_{m\to\infty}\mathcal{E}_m(u, v) = \lim_{m\to\infty} -\int_K\left(\sum_{x\in V_m\backslash V_0} v(x) f_m(x)\psi_x^{(m)}\right)\ \mathrm{d}\mu. = -\int_K v(y) f(y)\ \mathrm{d}\mu.$$

so $u \in \mathrm{dom}_\mu$

$$\left|\sum_{x\in V_m\setminus V_0} v(x)f_m(x)\psi_x^{(m)}(y) - v(y)f(y)\right|$$
$$\leq \left|\sum_{x\in V_m\setminus V_0} \big(v(x)f_m(x) - v(y)f(y)\big)\,\psi_x^{(m)}(y)\right| + \left|\sum_{x\in V_m\setminus V_0} v(y)f(y)\psi_x^{(m)}(y) - v(y)f(y)\right|$$

Looking at each term individually we see that

$$\left|\sum_{x\in V_m\setminus V_0} \big(v(x)f_m(x) - v(y)f(y)\big)\,\psi_x^{(m)}\right|$$
$$\leq \sum_{x\in V_m\setminus V_0}\Big(\big|v(x)f_m(x) - v(y)f_m(y)\big| + \big|v(y)f_m(y) - v(y)f(y)\big|\Big)\,\psi_x^{(m)}(y)$$

now again we will look at each term starting with $\big|v(y)f_m(y) - v(y)f(y)\big|$. Here

$$\big|v(y)f_m(y) - v(y)f(y)\big| = \big|v(y)\big|\big|f_m(y) - f(y)\big|$$

and since v is continuos on K, which is compact, there is a constant C_1 so $\big|v(y)\big| \leq C_1$ for all $x \in K$. Since f_m converges uniformly to f we can choose m_1 such that $\big|f_m(y) - f(y)\big| \leq \frac{\epsilon}{C_1}$ and the convergence of that term follows.

Next is $\big|v(x)f_m(x) - v(y)f_m(y)\big|$, here we find

$$\big|v(x)f_m(x) - v(y)f_m(y)\big| \leq \big|v(x)\big|\big|f_m(x) - f(y)\big| + \big|f_m(y)\big|\big|v(x) - v(y)\big|$$

First $\big|f_m(y)\big|\big|v(x) - v(y)\big|$, here $\big|f_m(y)\big| \leq C_2$ is uniformly bounded by some C_2 why?. By lemma 1.10 and continuity of v we can choose m_2 so $\big|v(x) - v(y)\big| \leq \frac{\epsilon}{C_2}$ for all $x \in F_{m_2}$. The reason we can choose m_2 is because $\psi_x^{(m)}$ is multiplied onto all the terms so we just restrict v to F_{m_2}.

Second $\big|v(x)\big|\big|f_m(x) - f(y)\big|$, again $\big|v(x)\big| \leq C_1$, but

$$\big|f_m(x) - f(y)\big| \leq \big|f_m(x) - f(x)\big| + \big|f(x) - f_m(y)\big| \leq \big|f_m(x) - f(x)\big| + \big|f(x) - f(y)\big| + \big|f(y) - f_m(y)\big|$$

First choose m_3 so $\big|f_{m_3}(x) - f(x)\big| \leq \frac{\epsilon}{3C_1}$, next by continuity of f choose m_4 so $\big|f(x) - f(y)\big| \leq \frac{\epsilon}{3C_1}$ for all $x \in F_{m_4}$, lastly choose m_5 so $\big|f(y) - f_{m_5}(y)\big| \leq \frac{\epsilon}{3C_1}$.

So we now have for $m \geq \max\{m_1, m_2, \ldots, m_5\}$ that

$$\left|\sum_{x\in V_m\setminus V_0} \big(v(x)f_m(x) - v(y)f(y)\big)\,\psi_x^{(m)}\right| \leq \epsilon \sum_{x\in V_m\setminus V_0} \psi_x^{(m)} \leq \epsilon \sum_{x\in V_m\setminus V_0} 1 \leq \epsilon.$$

We are now going to look at $\left|\sum_{x\in V_m\setminus V_0} v(y)f(y)\psi_x^{(m)}(y) - v(y)f(y)\right|$ here

$$\left|\sum_{x\in V_m\setminus V_0} v(y)f(y)\psi_x^{(m)}(y) - v(y)f(y)\right| = \big|v(y)f(y)\big|\left|\sum_{x\in V_m\setminus V_0} \psi_x^{(m)}(y) - 1\right|$$

Now to show $\left|\sum_{x\in V_m\setminus V_0} \psi_x^{(m)}(y) - 1\right| = 0$ as $m \to \infty$, we start by looking at $y \in V^*$. Choose m_6 so $y \in V_{m_6}$.

Next assume $y \in K$, then since V^* is dense in K we can choose a sequence $(x_n)_{n=1}$ where $x_n \to y$ as $n \to \infty$, and since $\psi_x^{(m)}$ is continuos on K then $\psi_x^{(m)}(x_n) \to \psi_x^{(m)}(y)$ as $n \to \infty$. This gives us the following

$$\left|\sum_{x\in V_m\setminus V_0} \psi_x^{(m)}(y) - 1\right| \leq \left|\sum_{x\in V_m\setminus V_0} \psi_x^{(m)}(y) - \sum_{x\in V_m\setminus V_0} \psi_x^{(m)}(x_n)\right| + \left|\sum_{x\in V_m\setminus V_0} \psi_x^{(m)}(x_n) - 1\right|$$

Now choose an n_1 so $\left|\psi_x^{(m)}(x_{n_1}) - \psi_x^{(m)}(y)\right| \leq \epsilon$ and an m_7 so $x_{n_1} \in V_{m_7}$. Both can then be bounded by $m \geq \max\{n_1, m_7\}$ $\square$

1.3 Iterated Function System

The iterated functions systems (IFS) gives us a way to work with self-similarly sets. Another reason for constructing the IFS is that we are able to prove that the Vicsek set actually exists.

Definition 1.12: A finite family of functions $\{\mathcal{S}_1, \mathcal{S}_2, \ldots, \mathcal{S}_m\}$ with $m \geq 2$ is called an *iterated function system* (or IFS) if exists $c_i \in (0,1)$ such that all $\mathcal{S}_i$ are *contractions*. Which is exactly

$$\left|\mathcal{S}_i(x) - \mathcal{S}_i(y)\right| \leq c_i |x-y| .$$

Further we call a non-empty compact subset K of X an *invariant set* for the IFS if

$$K = \bigcup_{i=1}^{m} \mathcal{S}_i(K) \tag{1.2}$$

Next we will show when there exists a unique invariant set.

For the proof we quickly define the Hausdorff metric

Definition 1.13: Define $A_\delta = \left\{x \in \mathcal{A} : \exists a \in A, |x-a| < \delta\right\}$ The *Hausdorff metric* is given by

$$d_H(X,Y) = \inf\{\delta : X \subset Y_\delta \text{ and } Y \subset X_\delta\}$$

The proof that d_H metric can be found in [Hen99].

Theorem 1.14: Consider the IFS given by $\{\mathcal{S}_1, \mathcal{S}_2, \ldots, \mathcal{S}_m\}$, then there is a unique invariante set K for the IFS.

Moreover, if we define a transformation S, on the class $\mathcal{A}$ of non-empty compact sets, by

$$S(E) = \bigcup_{i=1}^{m} \mathcal{S}_i(E), \quad \text{for } E \in \mathcal{A}$$

and write S^k to be the$k$'th iterate of S, so $S^0 = E$ and $S^k(E) = S(S^{k-1}(E))$ for $k \geq 1$. Then

$$F = \bigcap_{i=0}^{\infty} S^k(E)$$

for every set $E \in \mathcal{A}$ such that $S_i(E) \subset E$ for all i.

Proof. Let E be any set in $\mathcal{A}$ such that $S_i(E) \subset E$ for all i. By induction we find that $S^k(E) \subset S^{k-1}(E)$. Since $S^0 = E \supset S(E) = S^1(E)$, then $S^{k+1}(E) = S(S^k(E)) \subset S(S^{K-1}(E)) = S^k(E)$ by the induction assumption that $S^k \subset S^{k-1}$. So $S^k(E)$ is a decreasing sequence of non-empty compact sets, in which case it must have a non.empty compact intersection $K = \bigcap_{k\geq 0} S^k(E)$. Since $S^k(E)$ is decreasing, then $S(F) = F$ why? fulfilling (1.2).

Now moving on to uniqueness, We will need the following inequality

$$d_H(S(A), S(B)) = d_H\left(\bigcup_{i=1}^{m} \mathcal{S}_i(A), \bigcup_{i=1}^{m} \mathcal{S}_i(B)\right) \leq \max_{1\leq i\leq m} d_H(\mathcal{S}_i(A), \mathcal{S}_i(B)) \leq \max_{1\leq i\leq m} c_i \cdot d_H(A,B). \tag{1.3}$$

So assume A and B are both invariant sets, then $S(A) = A$ and $S(B) = B$, but that gives us that

$$d_H(A,B) = d_H(\mathcal{S}_i(A), \mathcal{S}_i(B)) \leq \max_{1\leq i\leq m} c_i \cdot d_H(A,B)$$

and since $0 < \max_{1\leq i\leq m} c_i < 1$, then $d_H(A,B) = 0$ so $A = B$ and it is unique.

Lastly we will prove (1.3). The first inequality comes by choosing $\delta = \max_{1\leq i\leq m} d_H(\mathcal{S}_i(A), \mathcal{S}_i(B))$ since that means that $\mathcal{S}_i(B) \subset (\mathcal{S}_i(A))_\delta$ for all $0 \leq i \leq m$, thus it will definitely hold for $\bigcup_{i=1}^m \mathcal{S}_i(B) \subset \left(\bigcup_{i=1}^m \mathcal{S}_i(B)\right)_\delta$ and vice versa. The second inequality comes from letting $\delta = d(A,B)$ then for every $a \in A$ we can choose some $b \in B$ such that $|a-b| \leq \delta$ in which case $\left|\mathcal{S}_i(a) - \mathcal{S}_i(b)\right| \leq c_i\delta$ and therefore $\mathcal{S}_i(A) \subset \left(\mathcal{S}_i(B)\right)_{c_i\delta}$. ☐

Now showing the existence of the Vicsek set is simply just checking that $\mathcal{S}_i$ are contractions. Which follows from

$$\left|\mathcal{S}_i(x) - \mathcal{S}_i(y)\right| = \left|\frac{1}{3}(x-p_i) + p_i - \left(\frac{1}{3}(y-p_i) + p_i\right)\right| = \frac{1}{3}|x-y| .$$

1.4 Something About Hausdorff and Walk Dimensions

What is hausdorff dimension? We first have to look at Hausdorff measure.

We say the countable collection of set $\{U_i\}_{i\geq 1}$ is a *δ-cover* of K if $K \subset \bigcup_{i\geq 1} U_i$ and $\operatorname{diam} U_i \leq \delta$. Here $\operatorname{diam}(A) := \sup_{x,y\in A} d(x,y)$ where $d(\cdot,\cdot)$ is a metric.

Definition 1.15: Let X be a metric space. If $K \subset X$ and $d \in [0,\infty)$ then let

$$H^d_\delta(K) = \inf\left\{\sum_{i=1}^\infty \operatorname{diam}(U_i)^d \ : \ \bigcup_{i=1}^\infty U_i \supseteq K, \operatorname{diam}(U_i) < \delta\right\}.$$

The d-dimensional *Hausdorff measure* is given by

$$\mathcal{H}^d(K) = \lim_{\delta\to 0} H^d_\delta(K)$$

The Hausdorff measure defined above, is not a measure i the normal measure theory sense but ratherit is an outer measure. Though it is possible to prove that the Hausdorff measure restricted to Borel-set is a measure.

Moving swiftly along, we now want to look at the behaviour of $\mathcal{H}^s$ as s changes. So let $U_{i i\geq 1}$ is a δ-cover of K then we have for $0 \geq s < t$ that

$$\sum_{i\geq 1} \operatorname{diam}(U_i)^t = \sum_{i\geq 1} \operatorname{diam}(U_i)^{t-s} \operatorname{diam}(U_i)^s \leq \delta^{t-s} \sum_{i\geq 1} \operatorname{diam}(U_i)^s.$$

Taking infima on both sides we get that $H^t_\delta(K) \leq \delta^{t-s} H^s_\delta(K)$. If we let $\delta \to 0$ we see that if $\mathcal{H}^s(K) < \infty$ then $\mathcal{H}^s(K) = 0$ since $\delta^{t-s} \to 0$. Likewise if $\mathcal{H}^t > 0$ then $\mathcal{H}^s = \infty$. T his tells us that if there is a point, say κ, such $\mathcal{H}^\kappa$ is both positive and finite, then $\mathcal{H}^s$ will be 0 for all $s > \kappa$ and ∞ for all $s < \kappa$. That such value is called the *Hausdorff Dimension* of K. Formally written:

Definition 1.16: The *Hausdorff dimension*, denoted $d_h(K)$ is given by

$$d_h(K) = \inf\left\{s \geq 0 : \mathcal{H}^s(K) = 0\right\} = \sup\left\{s \geq 0 : \mathcal{H}^s(K) = \infty\right\}.$$

When there is no confusion with regard to what set we are working with, we denote the Hausdorff dimension by d_h.

Now it is quite difficult to determine the Hausdorff dimension from the definition. Luckily we can with a few assumption find the Hausdorff dimension quite easily. For that theorem we need an important lemma.

Lemma 1.17: Let μ be a measure such that $\mu(F) \in (0,\infty)$ and suppose there are numbers $c, \epsilon > 0$ such that for some $s \geq 0$

$$\mu(U) \leq c \operatorname{diam}(U)^s$$

for all sets U with $\operatorname{diam}(U) \leq \epsilon$. Then $\mathcal{H}^s(F) \geq \frac{\mu(F)}{c}$ and $s \leq d_h$

Proof. If $\{U_i\}_{i\in I}$ is any cover of F then

$$0 < \mu(F) \leq \mu\left(\bigcup_{i\in I} U_i\right) \leq \sum_{i\in I} \mu(U_i) \leq c \sum_{i\in I} \operatorname{diam}(U_i)^s$$

by definition of measures and properties of μ. Taking infimum we get $\mathcal{H}^s_\delta \geq \frac{\mu(F)}{c}$ and therefore $\mathcal{H}^s \geq \frac{\mu(F)}{c} > 0$ so by definition 1.16 $s \leq d_h$. ☐

We first define the open set condition

Definition 1.18: We say that a IFS $\mathcal{S}_i$ satisfy the *open set condition* if there exists a non-empty bounded open set V such that

$$V \supset \bigcup_{i=1}^5 \mathcal{S}_i(V)$$

Theorem 1.19: Suppose definition 1.18 holds for the similarities $\mathcal{S}_i$ on $\mathbb{R}^n$ with ratios $0 < c_i < 1$ for $1 \leq i \leq m$. If there for K holds

$$K = \bigcup_{i=1}^{m} \mathcal{S}_i(K) \tag{1.4}$$

then $d_h = s$ where s is given by

$$\sum_{i=1}^{m} c_i^s = 1. \tag{1.5}$$

Moreover, for this value of s, $0 < \mathcal{H}^s(K) < \infty$

Proof. Let $\mathcal{I}_k$ denote the set of all k-term sequences $(i_1, \dots i_k)$ with $1 \leq i_j \leq m$. For any set A and any sequence $(i_1, \dots i_k) \in \mathcal{I}_k$ we denote $A_{i_1,\dots i_k} = \mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_k}(A)$.

We will start by finding an upper bound for the Hausdorff measure. We do this by finding an appropriate δ-cover for K. It follows that by repeated use of (1.4) that

$$K = \bigcup_{i=1}^{m} \mathcal{S}_i(K) = \bigcup_{i=1}^{m} \mathcal{S}_i \left(\bigcup_{j=1}^{m} \mathcal{S}_j(K) \right) = \bigcup_{i,j=1}^{m} \mathcal{S}_i \circ \mathcal{S}_j(K) = \dots = \bigcup_{\mathcal{I}_k} \mathcal{S}_{i_1,\dots i_k}(K) = \bigcup_{\mathcal{I}_k} K_{i_1,\dots i_k}.$$

Now noting that $\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_k}$ has the ratios $c_{i_1} \dots c_{i_k}$, we see

$$\sum_{\mathcal{I}_k} \operatorname{diam}(K_{i_1,\dots i_k})^s = \sum_{\mathcal{I}_k} (c_{i_1} \dots c_{i_k})^s \operatorname{diam}(K)^s = \left(\sum_{i_1} c_{i_1}^s \right) \dots \left(\sum_{i_k} c_{i_k}^s \right) \operatorname{diam}(K)^s = \operatorname{diam}(K)^s$$

where the last equality comes from (1.5). Now for any $\delta > 0$ we can choose k so $\operatorname{diam}(K_{i_1,\dots i_k}) \left(\max_{1 \leq i \leq m} c_i\right)^k \operatorname{diam}(K) \leq \delta$ since $\max_{1 \leq i \leq m} c_i \in (0,1)$. Thus $\left\{K_{i_1,\dots i_k}\right\}_{\mathcal{I}_k}$ is a δ-cover for K thus $\mathcal{H}^s_\delta(K) \leq \operatorname{diam}(K)^s$ in which case $\mathcal{H}^s(K) \leq \operatorname{diam}(K)^s$ and we have an upper bound.

Next we need a lower bound for the Hausdorff dimension, which is substantially more difficult.

Let $\mathcal{I} = \left\{(i_1, i_2, \dots) : 1 \leq i_j \leq m\right\}$ be the set of all infinite sequences and let $\mathcal{I}_{i_1,\dots i_k} = \left\{(i_1, \dots, i_k, q_{k+1}, \dots) : 1 \leq q_j \leq m\right\}$, so the first initial k "terms" are chosen and then the rest are the remaining permutations.

Now we create an outer measure μ on subset of $\mathcal{I}$ defined by $\mu(\mathcal{I}_{i_1,\dots i_k}) = (c_{i_1} \dots c_{i_k})^s$. From this outer measure μ we construct an outer measure on K named $\tilde{\mu}$ defined in the following way. For some subset $A \subseteq K$ -

$$\tilde{\mu}(A) = \mu\left(\left\{(i_1, i_2, \dots) \in \mathcal{I} : x_{i_1,i_2,\dots} \in A \cap K\right\}\right)$$

where

$$\left\{x_{i_1,i_2,\dots}\right\} = \bigcap_{k \geq 0} F_{i_1,\dots i_k}.$$

This is possible since S^k is decreasing as k increases. Proofs that μ and $\tilde{\mu}$ are indeed outer measures can be found in [Haz, Section 7].

Now we want to show that $\tilde{\mu}$ fulfills the properties of lemma 1.17.

Let V be the open set that satisfies definition 1.18. Since $\overline{V} \supset \mathcal{S}(\overline{V}) = \cup_{i=1}^m \mathcal{S}_i(\overline{V})$, then by theorem 1.14 the decreasing sequence $S^k(\overline{V})$ converges to K. In particularly we have $\overline{V} \supset S(\overline{V}) \supseteq \cup_{i=1}^m \mathcal{S}_i(\overline{V}) = F$ and $\overline{V}_{i_1,\dots i_k} = \mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_k}(\overline{V}) \supseteq \mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_k}(\overline{F}) = F_{i_1,\dots i_k}$

□

Chapter 2

And now for something completely different

In the previous chapter we constructed the Vicsek set, now we will consider the discrete Laplacian on this

Definition 2.1: Let $\ell(V_m)$ be the set of functions of $f : V_m \to \mathbb{R}$. Then for any $f \in \ell(V_m)$ the *discrete Laplacian* on V_m is defined as

$$\Delta_m f(p) = \text{CONST} \sum_{q \in V_{m,p}} (f(q) - f(p)), \quad p \in V_m \backslash V_0$$

Remark 2.2: One is able to describe the discrete Laplacian as a matrix. First let $p_1, \ldots p_{N_m}$ be the vertices in V_m then the *Laplacian matrix* is constructed in the following way:

$$L_{i,j} = \begin{cases} \# V_{m,p_i}, & \text{if } i = j \\ -1, & \text{if } i \neq j \text{ and } p_j \in V_{m,p_i} \\ 0, & \text{otherwise.} \end{cases}$$

Further $\# V_{m,p_i}$ is also called the *degree of* p_i

Now let $C(K)$ be the set of continuos functions on K. We then define

$$\mathcal{D} = \left\{ f \in C(K) \ : \ \exists g \in C(K), \lim_{m \to \infty} \max_{p \in V_m \backslash V_0} \left| \Delta_m f(p) - g(p) \right| = 0 \right\}$$

We will use the notation of $C_0(K)$ and D_0 to denote the sets of functions in $C(K)$ and D respectively which vanish on the boundary ∂K.

This set leads us to defining

Definition 2.3: The Kigami Laplacian $\Delta : \mathcal{D} \to C(K)$ is defined as

$$\Delta f(p) = g(p)$$

where g satisfies: $\lim_{m \to \infty} \max_{p \in V_m \backslash V_0} \left| \Delta_m f(p) - g(p) \right| = 0$

We will further consider the following discrete measure

$$\mu_m = \frac{1}{5^m \cdot 4 + 1} \sum_{p \in V_m} \delta_p$$

2.1 Discrete Fractional Gaussian Fields

Recalling remark 2.2 then we consider the eigenvalues $0 < \lambda_1^m \leq \lambda_2^m \leq \ldots \leq \lambda_{N_m}^m$ of $-\Delta_m$. Next let $(\Psi_i^m)_{1 \leq i \leq N_m}$ be the corresponding eigenfunctions with respect to the measure μ_m. This leads us to defining the following

Definition 2.4: Let $s \geq 0$ we define the *discrete Riesz kernel* on V_m as

$$G_s^m(x, y) = \sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \phi_i^m(x) \phi_i^m(y), \quad \text{for } x, y \in V_m$$

We can further by construction see that the matrix $(G_s^m(x,y))_{x,y\in V_m}$ is symmetric and is non-negative Why?, thus it can be the covariance matrix of a Gaussian vector.

Theorem 2.5 (Bochner-Minlos)**:** Assume $\mathbb{X}$ is a nuclear space. Any continuos positive-definite linear functional Λ on $\mathbb{X}$ satisfying $\lambda(0)=1$ is the Fourier transform of a countably additive positive normalized measure on $\mathbb{X}'$ REFERENCE IS FROM WIKIPEDIA Ref is here

Theorem 2.6: Let $s\geq 0$, there exists a centered Gaussian distribution X_s on $S'(K)$ such that for $f,g\in S(K)$

$$\mathbb{E}[X_s(f)X_s(g)] = \int_K (-\Delta)^{-s} f (-\Delta)^{-s} g \, \mathrm{d}\mu$$

Proof. The space $S(K)$ is a nuclear space WHY, it is enough to use Bochner-Minlos theorem 2.5 on the functional

$$\Lambda : f \to \exp\left(-\frac{1}{2}\|f\|_k^2\right).$$

DEFINE NORM

This is enough since theorem 2.5 then gives an unique probability measure ν such

$$\exp\left(-\frac{1}{2}\|f\|_k^2\right) = \int_{S'(K)} e^{i\langle x,f\rangle} \, \mathrm{d}\nu(x).$$

Notice that the right side is an characteristic function for some random variable. If we choose $f=t\cdot g$ for some $g\in S(K)$ and $t\in\mathbb{R}$ Where is t from?(notice that $t\cdot g\in S(K)$) then the left side of the equation becomes the characteristic function for the normal distribution with mean zero and variance $\|f\|_s^2$. Thus we have the X_s we wanted.

Now to prove that Λ is satisfies the requirements of theorem 2.5. Obviously $\Lambda(0)=1$, further it follows from the fact that $\|t\cdot f\|_k^2$ is an inner product that withTHEOREM: 2.4 A SURVEY Λ is positive definite. Lastly we need to prove it is continuous at zero.

First we note that since $(-\Delta)^{-s}$ is a bounded operator we have that there exists some constant $C>0$ so $\left\|(-\Delta)^{-s} f\right\|_{L^2(K,\mu)} \leq C\|f\|_{L^2(K,\mu)}$.

Now let $\epsilon>0$ be given, choose

$$\delta = \begin{cases} \left(\frac{-2\cdot\ln(1-\epsilon)}{C}\right)^{\frac{1}{2}}, & \text{when } \epsilon\in(0,1) \\ 1, & \text{when } \epsilon\geq 1. \end{cases}$$

First if $\epsilon\geq 1$ notice that

$$\left|\exp\left(-\frac{1}{2}\|f\|_s^2\right) - \exp\left(-\frac{1}{2}\|0\|_s^2\right)\right| \leq 1 \leq \epsilon, \quad \forall f\in S(K)$$

in which we can choose δ to be any real number larger than 0. Moving on to $\epsilon\in(0,1)$ we get

$$\begin{aligned}
\left|\exp\left(-\frac{1}{2}\|f\|_s^2\right) - \exp\left(-\frac{1}{2}\|0\|_s^2\right)\right| &\leq \left|\exp\left(-\frac{1}{2}C\|f\|_{L^2(K,\mu)}\right) - 1\right| \\
&= \left|\exp\left(-\frac{1}{2}C\int_K |f|^2 \, \mathrm{d}\mu\right) - 1\right| \\
&\leq \left|\exp\left(-\frac{1}{2}C\int_K \delta^2 \, \mathrm{d}\mu\right) - 1\right| \\
&= \left|\exp\left(-\frac{1}{2}C\int_K \frac{-2\cdot\ln(1-\epsilon)}{C} \, \mathrm{d}\mu\right) - 1\right| \\
&= \left|\exp\left(\ln(1-\epsilon)\right) - 1\right| \\
&= \epsilon
\end{aligned}$$

in the third to last equality using that μ is a probability measure. In which case we have continuity in 0 as desired. $\square$

Chapter 3

Main Theorem

3.1 Something

We have now proven the existence of the gaussian fields and are now ready to prove the convergence.

Theorem 3.1: Let $s \geq 0$, then X_s^m converges to X_s in distribution in $S'(K)$ when $m \to \infty$

Proof. First since $S(K)$ is a nuclear space, by [BDW17, Corollary 2.4] it is enough to prove the convergence of the characteristic functional, e.i that for all $f \in S(K)$

$$\mathbb{E}\left[\exp(itX_s^m(f_m))\right] \to \mathbb{E}\left[\exp(itX_s(f))\right], \quad \text{as } m \to \infty$$

where $f_m = f\mid_{V_m}$. It follows fra INSERT REF that

$$\mathbb{E}\left[\exp(itX_s^m(f_m))\right] = \exp\left(-\frac{1}{2}t^2\mathbb{E}\left[(X_s^m(f_m))^2\right]\right) = \exp\left(-\frac{1}{2a_m}t^2\sum_{p\in V_m} f_m(p)(-\Delta_m)^{-2s}f_m(p)\right)$$

Likewise we find that via coll. 2.11 + 2.9 we get that

$$\mathbb{E}\left[\exp(itX_s(f))\right] = \exp\left(-\frac{1}{2}t^2\mathbb{E}\left[(X_s(f))^2\right]\right) = \exp\left(-\frac{1}{2}t^2\int_K f(-\Delta)^{-2s}f\ \mathrm{d}\mu\right).$$

Where we notice that

$$\begin{aligned}\mathbb{E}\left[(X_s(f))^2\right] &= \int_K (-\Delta)^s f(-\Delta)^s f\ \mathrm{d}\mu \\ &= \int_K\int_K f(p)f(q)G_{2s}(p,q)\ \mathrm{d}\mu(p)\ \mathrm{d}\mu(q) \\ &= \int_K f(-\Delta)^{-2s}f\ \mathrm{d}\mu.\end{aligned}$$

The proof now follows from lemma 3.2. □

Lemma 3.2: For all $f \in S(K)$ and $s \geq 0$ the following holds

$$\frac{1}{a_m}\sum_{p\in V_m} f_m(p)(-\Delta_m)^{-2s}f_m(p) \to \int_K f(x)(-\Delta)^{-2s}f(x)\ \mathrm{d}\mu(x), \quad \text{as } m \to \infty$$

Proof. For $s = 0$ we find that

$$\begin{aligned}(-\Delta_m)^{-0}f &= \sum_{i=1}^{N_m}\frac{1}{(\lambda_i^m)^0}\phi_i^m(x)\frac{1}{a_m}\sum_{y\in V_m}\psi_i^m(y)f(y) \\ &= \sum_{i=1}^{N_m}\phi_i^m(x)\int_{V_m}\phi_i^m(y)f(y)\ \mathrm{d}\mu_m \\ &= f_m.\end{aligned}$$

Likewise by definition of $(-\Delta)^{-s}$ we find

$$(-\Delta)^{-0} f = \sum_{i=1}^{\infty} \phi_i \int_K \phi_i(y) f(y) \, \mathrm{d}\mu = f.$$

and the lemma follows from REF!!!. Now assume $s > 0$ then by Fubini's theorem and Some random ass fact we have that

$$\frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m f_m(p) (-\Delta_m)^{-2s} f_m(p) = \frac{1}{\Gamma(2s)} \int_0^{\infty} t^{2s-1} \frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m f_m(p).$$

Now by B.2.7 in [Kig01] and the definition of f_m we find that INSERT CALCULATION HERE

$$\begin{aligned} \frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m f_m(p) &= \frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m \sum_{j=1}^{\infty} \langle f_m, \phi_i \rangle \phi_i \\ &= \frac{1}{a_m} \sum_{p \in V_m} f_m(p) \sum_{j=1}^{\infty} \langle f_m, \phi_j \rangle P_t^m \phi_j \\ &= \frac{1}{a_m} \sum_{p \in V_m} f_m(p) \sum_{j=1}^{\infty} \langle f_m, \phi_j \rangle e^{-\lambda_j^m t} \phi_j \\ &\leq e^{-\lambda_1^m t} \frac{1}{a_m} \sum_{p \in V_m} f_m(p)^2 \end{aligned}$$

where we can guarantee the inequality for all j since $0 < \lambda_1 < \lambda_2 < .\lambda_3 < \ldots$

Note that $\sup_m \frac{1}{a_m} \sum_{p \in V_m} f_m(p)^2 < \infty$ since $f \in S(K) \subset C(K)$ so f is bounded and $\# V_m < \infty$ for all m. We can now by dominated convergence, Fubini's theorem and lemma 3.3 find that

$$\begin{aligned} \lim_{m \to \infty} \frac{1}{a_m} \sum_{p \in V_m} f_m(p) (-\Delta_m)^{-2s} f_m(p) &= \frac{1}{\Gamma(2s)} \int_0^{\infty} t^{2s-1} t^{2s-1} \lim_{m \to \infty} \frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m f_m(p) \, \mathrm{d}t \\ &= \frac{1}{\Gamma(2s)} \int_0^{\infty} t^{2s-1} \int_K f(x) P_t f(x) \, \mathrm{d}\mu(x) \, \mathrm{d}t \\ &= \frac{1}{\Gamma(2s)} \int_K f(x) \int_0^{\infty} t^{2s-1} P_t f(x) \, \mathrm{d}t \, \mathrm{d}\mu(x) \\ &= \int_K f(x) (-\Delta)^{-2s} f(x) \, \mathrm{d}\mu(x) \end{aligned}$$

as desired. □

In the following proof we will use results from [Kur69]. Before stating the lemma and proof we will give some arguments as to why we are allowed to use these results.

Firstly we note that since Δ is a bounded operator the by [Paz92, Theorem 1.2] it is the infinitesimal generator of a uniformly continuos semigroup.

Secondly we have that the sequence of laplacians $\{\Delta_m\}_{m \geq 0}$ indeed coincides with extended limit defined in [Kur69, page 355]. This follows from theorem 1.11.

Lastly the range $\mathcal{R}(\lambda_0 - \Delta)$ is dense in L^2 why?

Lemma 3.3: For all $f \in S(K)$ and $t > 0$,

$$\lim_{m \to \infty} \frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m f_m(p) = \int_K f(x) P_t f(x) \, \mathrm{d}\mu(x)$$

where $f_m = f\,|_{V_m}$.

Proof. In this proof we wish to use [Kur69, theorem 2.1] so recall that by the definition of the Laplacian on K we have that

$$\lim_{m \to \infty} \sup_{p \in V_m} \left| \Delta_m f_m(p) - \Delta f(p) \right|.$$

Thus by [Kur69, theorem 2.1] we have that for all $t \geq 0$

$$\lim_{m\to\infty} \sup_{p\in V_m} \left| P_t^m f_m(p) - P_t f(p) \right| = 0. \tag{3.1}$$

So if we now write

$$\frac{1}{a_m} \sum_{p\in V_m} f_m(p) P_t^m f_m(p) = \int_{V_m} f_m P_t^m f_m \ \mathrm{d}\mu_m.$$

Thus

$$\int_{V_m} f_m P_t^m f_m \ \mathrm{d}\mu_m = \int_{V_m} f_m \left(P_t^m f_m - (P_t f)_m \right) \ \mathrm{d}\mu_m + \int_{V_m} f_m (P_t f)_m \ \mathrm{d}\mu_m.$$

The first integral converges to 0 as $m \to \infty$ because of (3.1). Further by lemma we ave $\int_{V_m} f_m (P_t f)_m \ \mathrm{d}\mu_m \to \int_K f P_t f \ \mathrm{d}\mu$ as $m \to \infty$. □

Lemma 3.4: The sequence of measure $(\mu_m)_{m\geq 0}$ defined in REF!!! converges to the normalized Hausdorff measure μ on the Vicsek set K in the weak topology. That is

$$\lim_{m\to\infty} \int_K f \ \mathrm{d}\mu_m = \int_K f \ \mathrm{d}\mu$$

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